

Herd Immunity and Learning Externalities in Markets of Shared Foundation Models

Vignesh Nagarajan
Texas A&M University
College Station, TX, USA
vigneshn26@tamu.edu

Shriraghav Ashok
University of California, Berkeley
Berkeley, CA, USA
sashok24@berkeley.edu

Abstract—Firms that price, lend and rank with machine learning retrain on data their own models helped produce. When many of them fine-tune the same foundation model, one firm’s retraining changes what every other firm learns from next, and no firm accounts for it. We model this as a market of retraining firms and show that it is stable exactly when one number, the feedback reproduction number, stays below one. That number depends on how many independent models sit under the market, not on how many firms there are: fifty firms built on one vendor’s model behave like one learner fifty times larger. Three remedies follow. Retraining in smaller steps helps only below a critical level of crowding. Correcting for feedback works like vaccination, and the share of firms that must correct is the epidemiological coverage formula for an imperfect vaccine, including an efficacy below which no coverage is enough. A tax on aggressive adaptation closes the gap between private and social incentives, and without it every market of two or more firms over-adapts. In an order-flow market simulator, the base amplification law reproduces published runs exactly.

Index Terms—learning externalities, algorithmic monoculture, foundation models, systemic risk, herd immunity, performative prediction, Pigouvian taxation

I. INTRODUCTION

Picture fifty dealers quoting the same bonds. Each one retrains its pricing model on the order flow it saw since the last retraining, and each one, tested on its own, is fine: replay its own history and it settles down. Put the fifty in one market and the market can still blow up. The flow a dealer sees depends on how the other forty-nine priced, so every retraining changes what everyone else learns from next. None of the fifty models accounts for that.

We call this a *learning externality*. The broader idea, that sensible individual behavior can add up to collective fragility, is old. Banks that each diversify optimally end up holding the same portfolios and fail together [1], [2]. Quantitative funds running similar strategies unwound together in August 2007, and no single fund’s risk report showed the crowding [3]. What is new is that the agents now learn, and more and more of them learn from the same place. Firms that fine-tune one vendor’s foundation model make correlated errors [4]. We show that they also push the market in correlated directions, and that this, rather than the number of firms, is what decides whether the market is stable.

The tool people use today looks at one model at a time. A learner whose deployment reshapes its own data converges

under repeated retraining when a single number, its performative modulus m , stays below one [5]. Certify $m < 1$ and the loop is safe. In a market that certificate is not enough. The quantity that matters is a market-level number we call the *feedback reproduction number* m_N , the factor by which a shared disturbance grows per round of retraining. The market converges exactly when $m_N < 1$, and m_N depends on how many independent models sit under the market. Fifty firms built on one vendor’s model behave like one learner fifty times larger. Fifty firms whose models push the market in unrelated directions behave like one firm.

The result we like best started as a mistake. Our first version of the herd-immunity result took a limit in which correction is perfect, and it produced a clean threshold: correct enough firms and the market is safe. When we checked that threshold against the exact spectral radius on random markets, it was wrong in the dangerous direction. It called 11.8% of configurations stable that are unstable at every finite correction strength. Working out the exact threshold turned up a formula epidemiologists already use, the coverage a population needs when the vaccine is imperfect [6], [7], together with an efficacy below which no coverage is enough. The clean limit is still in the paper, but as a corollary with its error direction stated.

Contributions

- 1) *Effective crowding*. The stability of a market of retraining firms is governed by $m_N = N_{\text{eff}}m_1$, where N_{eff} depends on the leading eigenvalue of a correlation matrix of the firms’ response directions (Section IV). Diversity indices built on mean similarity can call an unstable market safe.
- 2) *Herd immunity*. An exact stability threshold for a market mixing corrected and uncorrected firms, which reduces to the imperfect-vaccine coverage law and implies a critical efficacy. Correction and model diversity substitute for each other along a frontier we compute (Section V).
- 3) *Cadence and a tax*. Retraining in smaller steps helps only below a critical crowding level, and a Pigouvian fee prices the externality. Without the fee, every market of two or more firms adapts too aggressively (Section VI).
- 4) *Labeled evidence*. One numerical check is measured in an order-flow market. The rest confirm the formulas against the model they came from, and each one says

which kind it is. Full proofs, code and 542 numerical certificates are public [8].

What this paper is not

It is not an empirical study of a deployed market. Our one measured result comes from an order-flow simulator we built for earlier work [9]. We have not yet measured the key eigenvalue on any real set of deployed models, and Section VIII says how we would. What we offer is the quantity a supervisor would need to measure and the three instruments that act on it.

II. RELATED WORK

A. Collective behavior and contagion

Threshold models explain how a behavior spreads once enough others have adopted it [10], and informational cascades explain why people rationally copy their predecessors and why the resulting consensus is fragile [11]. Both are about people imitating people. In our setting nobody imitates anyone. Firms become correlated because they learn from shared data through shared models. The epidemiological link is an identity rather than an analogy: R_0 is defined as the spectral radius of a next-generation operator [12], m_N is the spectral radius of a linearized retraining operator, and so the two thresholds are the same statement. That is why the vaccination results carry over with their refinements, including the imperfect-vaccine law. Heterogeneous contact structure famously changes epidemic thresholds [13]. Here the analogous structure is the alignment matrix.

B. Algorithmic monoculture

When many firms adopt one algorithm, welfare can fall even if that algorithm is the better one, because independent errors stop cancelling [14]. Shared models also concentrate harm on the same individuals [15], and a shared evaluation algorithm changes outcomes in matching markets [16]. These results describe a market at rest. None gives a stability boundary, so a market can look fine at equilibrium and still be one retraining away from diverging.

C. Performative prediction with many learners

Performative prediction studies learners whose deployments shift their own data [5]. With several learners in one environment, convergence conditions exist that bound scalar sensitivities against curvature [17], and in the fully aligned case, increasing learning rates and agent counts can drive the dynamics into chaos [18]. Our condition bounds a spectral radius instead, and Proposition 1 shows two markets that agree on every scalar such a criterion reads but differ in stability. We also compose two single-learner tools with the market: lazy deployment, where a learner takes a few gradient steps per round [19], and feedback-aware updates that correct for the learner's own influence [20], [21].

D. Systemic risk and random matrices

The largest eigenvalue of a correlation matrix as the weight of a common market mode is standard in econophysics [22], [23]. We compute it from a different matrix, response directions instead of returns, and it enters a stability condition rather than a portfolio. Reading economic conditions from public text works through an exogenous channel [24]. The supervisory estimator in Section VIII reads an endogenous one, and its signal gets stronger as the market approaches the boundary it is watching for.

III. A MARKET OF RETRAINING FIRMS

A. The symmetric market

Start with N dealers quoting the same instruments and drawing on one pool of informed order flow. When one dealer tightens its quotes, more toxic flow reaches every dealer, in proportion to a spillover $\kappa \in [0, 1]$. Each dealer retrains periodically on the flow it has seen, and on its own it is stable exactly when its performative modulus satisfies $m_1 < 1$. Here $m_1 = \varepsilon\beta/\gamma$, where ε is how strongly the flow responds to the dealer's quotes, β is how sensitive its loss is to that response, and γ is the curvature of its objective [5].

Linearize the joint retraining map around the market's equilibrium. Its Jacobian is $J = -m_1[(1 - \kappa)I + \kappa\mathbf{1}\mathbf{1}^\top]$, and its largest mode, the one where all dealers move together, has size $m_1(1 + \kappa(N - 1))$. The market therefore goes unstable at a response strength $1 + \kappa(N - 1)$ times smaller than any single dealer can tolerate. This law comes from our earlier work [9], and Section VII measures it. Everything below replaces one object in it: the all-ones matrix $\mathbf{1}\mathbf{1}^\top$, which says every dealer pushes the flow in the same direction.

The economics is worth a sentence. A price change is a transfer between parties. Changing the data a competitor learns from changes its forecasting technology, which is a real efficiency loss, a technological externality in the classical sense [25]. And nothing in dealer A's validation data contains dealer B, so no single-model test can see it.

B. Response directions and the alignment matrix

In a real market firms do not all push the same way. Give firm i a response Jacobian $E_i \in \mathbb{R}^{d \times d}$, describing how its deployment reshapes the flow it faces. The *alignment matrix* R has entries

$$r_{ij} = \frac{\langle \text{vec } E_i, \text{vec } E_j \rangle}{\|E_i\|_F \|E_j\|_F}, \quad (1)$$

the correlation between two firms' feedback directions. It is positive semidefinite with ones on the diagonal, like a correlation matrix of returns, but it measures something different: whether two firms' models distort the market the same way.

We work under four standing assumptions. (A1) Stability statements linearize the joint map at equilibrium, and the simulator serves as the nonlinear check. (A2) Firms have equal moduli $m_i = m_1$; unequal moduli give two-sided bounds. (A3) Firms retrain on a common clock. (A4) Firms couple through one pool with scalar spillover κ .

Lemma 1 (Reduction). *Suppose each firm’s deviation from equilibrium is a scalar x_i along its own unit response direction, and all $\|E_i\|_F$ equal ε . Each firm feels its own distortion in full and a competitor’s with weight κ , both measured along its own direction. Each firm best-responds under a γ -strongly convex objective with sensitivity β . Then the joint retraining map is linear with Jacobian*

$$J = -m_1[(1 - \kappa)I + \kappa R]. \quad (2)$$

The proof is two lines. The pool carries the total distortion $\sum_j x_j E_j$, firm i reads it along its own direction, and the inner products are εr_{ij} . The hypothesis that firms read the pool only along their own direction is literal in a dealer market. A dealer sees order flow only where it quotes. Flow that a competitor’s repricing pushes into instruments the dealer does not trade never reaches its training set, however large it is. That is why two firms perturbing the market in orthogonal directions do not affect each other, and why R shows up at all.

IV. EFFECTIVE CROWDING

Counting firms gives the wrong answer. The right count is the leading eigenvalue $\lambda_{\max}(R)$, which runs from 1, when all response directions are orthogonal, to N , when they all coincide.

Theorem 1 (Alignment sets the reproduction number). *Under (A1)–(A4) and Lemma 1, the market’s spectral radius is*

$$m_N = N_{\text{eff}} m_1, \quad N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1), \quad (3)$$

and the market is stable if and only if $m_N < 1$.

In words: a market behaves like N_{eff} copies of one firm, and N_{eff} counts independent models, not firms.

Four cases make this concrete. If every firm shares one model, $R = \mathbf{1}\mathbf{1}^\top$ and we recover the symmetric law $N_{\text{eff}} = 1 + \kappa(N - 1)$. If every firm’s direction is orthogonal to every other, $R = I$ and $N_{\text{eff}} = 1$: a hundred firms pushing in a hundred unrelated directions are dynamically one firm. If each firm’s response is a share s from a common foundation model, vendor or pretraining corpus and $1 - s$ its own, alignments concentrate at s and $N_{\text{eff}} \rightarrow 1 + \kappa s(N - 1)$. And in every case $N_{\text{eff}} \geq 1$, so interaction never makes a market more stable than its members are alone.

Proposition 1 (Scalars are not enough). *Fix $N \geq 2$, $\kappa \in (0, 1]$ and $1/(1 + \kappa(N - 1)) < m_1 < 1$. The markets with $R = I$ and $R = \mathbf{1}\mathbf{1}^\top$ share ε , β , γ , m_1 and κ , yet the first is stable and the second is not.*

At $N = 20$, $\kappa = 0.8$ and $m_1 = 0.15$, the orthogonal market has $m_N = 0.15$ and the monoculture has $m_N = 2.43$. One converges and the other diverges, and no Lipschitz constant or curvature bound in either model tells them apart. The alignment matrix does.

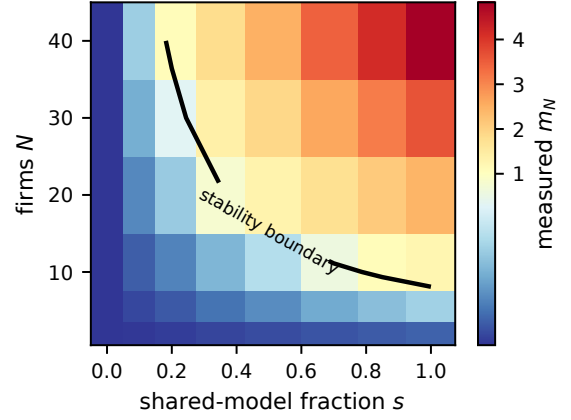


Fig. 1. Dry run in the reference environment: m_N across 48 cells of firm count N and shared-model fraction s , with the predicted stability boundary. It agrees with Theorem 1 in every cell to within 7.1×10^{-15} (Section VII).

A. Provenance, not concentration

The supply-chain form $m_N = m_1(1 + \kappa s(N - 1))$ says the stability condition is driven by where the models come from. A competition regulator looks at concentration. Fifty equal-share firms have the lowest possible Herfindahl index [26] and look perfectly competitive. If all fifty fine-tune one vendor’s model, they are dynamically a monoculture. Error correlation across more than 350 hosted language models is substantial and clusters within providers [4], so the premise of the decomposition is already visible in deployed systems. Fig. 1 shows the stability boundary in the (N, s) plane.

B. Why mean similarity fails

A natural diversity index averages pairwise similarity. The average understates *clustered* alignment. Take ten firms, three tightly aligned and the rest orthogonal, with $\kappa = 0.8$. Then $N_{\text{eff}} = 2.60$, against 1.48 for a uniform configuration with the same mean. At $m_1 = 0.5$ the clustered market has $m_N = 1.30$ and diverges, while the mean-based index reports 0.74 and calls it safe with room to spare. The error only ever runs one way: $N_{\text{eff}} \geq 1 + \kappa(N - 1)\bar{r}$ for every R , so a mean-based index never overstates risk. It can only miss it.

V. HERD IMMUNITY

Slowing down is something a firm can do on its own (Section VI). Correction is different: it only works if enough firms do it. A *corrected* firm runs a feedback-aware update that accounts for its own influence on the data [20], [21]. Its modulus shrinks by a factor $\theta = \gamma/\gamma_{\text{PO}} < 1$, where γ_{PO} is the curvature of the objective once the feedback is priced in. Correction changes how hard a firm pushes, not which way, so R is unchanged.

Consider a market of N_b uncorrected (“blind”) firms and $N_c = N - N_b$ corrected ones, with corrected fraction $\rho_c = N_c/N$.

Theorem 2 (Mixed-market stability). *Under (A1)–(A4) with supply-chain alignment s and $1 \leq N_b \leq N-1$, write $\tilde{\kappa} = \kappa s$. The spectral radius is the larger root*

$$\rho(J) = \frac{1}{2} \left(P + \sqrt{P^2 - 4Q} \right), \quad (4)$$

where $P = (1 - \tilde{\kappa})m_1(1 + \theta) + \tilde{\kappa}m_1(N_b + \theta N_c)$ and $Q = \theta m_1^2(1 - \tilde{\kappa})N_{\text{eff}}$.

The radius comes from a secular equation, a diagonal matrix plus a rank-one term, which with two correction levels collapses to a quadratic. N_{eff} sits inside Q , so the mixed market inherits the crowding of Section IV directly.

A. The limit that was too optimistic

Let correction become perfect, $\gamma_{\text{PO}} \rightarrow \infty$. Then only blind firms can sustain an unstable cycle, and the market is stable when $N_b < 1 + (1/m_1 - 1)/(\kappa s)$. This is the clean threshold we started with, and it errs in the wrong direction. The radius is nondecreasing in θ (Perron–Frobenius, on a nonnegative matrix), so the perfect-correction limit understates it. On random draws it calls 2157 of 18313 configurations stable that are unstable at every finite correction strength, 11.8% with exact 95% interval [11.3%, 12.3%]. The same monotonicity gives the reassuring half: correcting a firm never destabilizes the market.

B. The imperfect-vaccine law

In the corner $\kappa = s = 1$ the perfect-correction limit reads $\rho_c^* = 1 - 1/m_N$. This is exactly the herd-immunity threshold $1 - 1/R_0$, with m_N playing the role of the reproduction number. A market of ten firms at $m_N = 2.5$ needs six of them corrected, if correction is perfect. It is not, and at the same corner the exact condition solves to

$$\rho_c^*(e) = \frac{1 - 1/m_N}{e}, \quad e = 1 - \frac{\gamma}{\gamma_{\text{PO}}}, \quad (5)$$

where e is the *efficacy* of correction. This is the standard coverage requirement for an imperfect vaccine [6], [7]. The clean law assumes homogeneous mixing, and the corner $\kappa = s = 1$ is the market’s version of that assumption.

Corollary 1 (Critical efficacy). *No corrected fraction stabilizes the market when $\theta > 1/m_N$. Correction is a usable lever only if $\gamma_{\text{PO}}/\gamma > m_N$.*

At $m_N = 2.5$, correction must improve curvature by more than a factor of 2.5 or it cannot save the market at any coverage. Fig. 2 shows the thresholds moving right as efficacy falls.

C. Diversity as a substitute

The required coverage ρ_c^* rises with s wherever it is positive, so a market can reach stability along either axis. At $N = 20$, $m_1 = 0.15$ and $\kappa = 0.8$, a monoculture needs 12 corrected firms, a market at $s = 0.5$ needs five, and a market at $s = 0.2$ needs none.

Both remedies are public goods. A corrected firm pays for its correction and everyone shares the stability. A firm

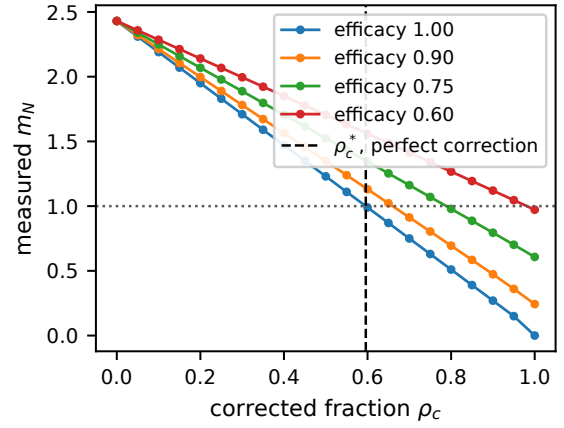


Fig. 2. Dry run: market modulus m_N against corrected fraction at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$, $s = 1$, for four efficacies. The dashed line is the perfect-correction threshold. The gap between it and each crossing is the error Section V describes, not noise.

that picks an idiosyncratic model pays for that choice and supplies stability to competitors for free. So the market under-supplies both [27], and the trade-off between them is a trade-off between two goods that are each too scarce.

VI. TWO MORE LEVERS: CADENCE AND A TAX

A. Moving less per round

The one lever a firm can pull alone is to change less each time it retrains. A learner that takes only K gradient steps per deployment has outer slope $-m + c^K(1 + m)$, where $c = 1 - \eta\gamma$ is its per-step contraction [19]. Because c depends only on the firm’s own step size and curvature, it does not change with the number of competitors, and that is what lets the single-firm result compose with the market.

Theorem 3 (Cadence window). *With synchronous K -step retraining, the joint deployment map is $c^K I + (1 - c^K)J$, and the market is stable if and only if*

$$K < K_{\max} = \frac{\ln((m_N - 1)/(m_N + 1))}{\ln c}, \quad (6)$$

with $K_{\max} = \infty$ when $m_N \leq 1$.

The window shrinks as crowding grows, and it can close. Since $K \geq 1$, a stabilizing cadence exists at all only if $m_N < (1 + c)/(1 - c)$, which is 9 at $c = 0.8$. Past that critical crowding level, no step budget is small enough.

Crowding comes from provenance, so a firm’s step budget depends on its competitors’ choice of vendor. At $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$ and $c = 0.8$, the window is 20.68 steps at $s = 0.25$, 5.28 at $s = 0.5$ and 2.53 at $s = 1$. Nobody entered the market, yet every incumbent lost most of its step budget. The price of a smaller K is staleness, and it is the same number: after a round, the gap between a firm’s deployed model and its best response is exactly c^K times what it was.

B. Pricing the externality

Inside the stable region the market's common mode behaves like an autoregressive process with coefficient $-m_N$, so its stationary variance is $V(m_N) = \sigma^2/(1 - m_N^2)$. We use V as a smooth stand-in for the risk of divergence. It blows up exactly at the boundary, and nothing below needs more than V being positive, increasing and convex.

Each firm chooses an adaptation aggressiveness a_i , meaning its cadence, learning rate or responsiveness, and gets a concave benefit $B(a_i)$ through its modulus $m_i = \mu(a_i)$. Firm i puts weight w_i on common-mode variance and a planner puts weight $W > \sum_i w_i$, because clients also bear execution-quality variance that no dealer's objective includes. The results depend only on the sign of that gap, not its size.

A firm's marginal effect on the market is $\partial m_N / \partial m_i = N_{\text{eff}} v_i^2$, where v is the leading eigenvector of the coupling matrix. These shares add up to N_{eff} , not to one. On exchangeable R each is N_{eff}/N , so a naive commons guess of $1/N$ is off by a factor of N_{eff} , which is 16.2 at $N = 20$, $\kappa = 0.8$ and $s = 1$.

Theorem 4 (The wedge). *At a symmetric profile, the private and social first-order conditions differ only in the price of crowding. The marginal cost a firm ignores is*

$$t^* = (W - w_i) V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a), \quad (7)$$

where $V'(m_N) = 2\sigma^2 m_N / (1 - m_N^2)^2$, and a per-unit fee t^* on aggressiveness makes the decentralized equilibrium coincide with the symmetric social optimum.

The fee rises with N , κ , s and m_N , and diverges like $(1 - m_N)^{-2}$ at the boundary. That rate is the part a regulator should worry about: a fee set when the market looked calm and then left alone will be far too low by the time the market is close to the edge.

Corollary 2 (Over-adaptation). *The untaxed equilibrium adapts more aggressively than the social optimum for every $N \geq 2$, even with no client exposure. At $N = 1$ with no client exposure the wedge is exactly zero.*

The same fee also prices shared-model adoption. Since $\partial m_N / \partial s = m_1 \kappa (N - 1)$, the provenance channel grows linearly in N and does not decay, while a firm's private gain from adopting the shared model does not grow with market size. Left alone, adoption drifts toward the unstable configuration. That is why a diversity floor is a separate instrument from a cadence cap and not a restatement of it. The three levers are quantity regulation, a technology mandate and a price, and choosing among them under uncertainty is Weitzman's question [28]. We supply the objects needed to ask it.

VII. EVIDENCE

The checks in Table I are not all the same kind of evidence, and each row says which kind it is. *Measured* means the run took place in an order-flow simulator: a real N -dealer market

TABLE I
THE SIX CHECKS AND THEIR STATUS

Check	Tests	Status	Outcome
Amplification	Sec. III	measured	$1.7428\times, 3.1567\times$
Phase diagram	Thm. 1	dry run	48 cells, error $\leq 7.1\text{e-}15$
Clustered firms	Thm. 1	dry run	$m_N = 1.30$; index 0.74
Cadence window	Thm. 3	dry run	175 cells, 0 disagree
Herd immunity	Thm. 2	dry run	12/14/16/20 firms
Substitution	Thm. 2	dry run	18 of 18 points
Over-adaptation	Thm. 4	dry run	12 configs, 0 contradict

with informed flow, spreads and inventory, which reduces exactly to the single-dealer market at $N = 1$. *Dry run* means the run took place in a linearized reference environment that implements the model's response geometry and nothing else. A dry run checks that a formula was derived correctly from the model it claims to describe, and these checks have caught real errors, including the one in Section V. A dry run cannot show that a real market behaves like the model. Every closed form in the paper has a numerical certificate that fails loudly on any disagreement, and all 542 pass [8].

A. The measured check

We reran the shared-pool market from our earlier work [9] and measured common-mode moduli of 0.7856, 1.3692 and 2.4799 at $N = 1, 2, 3$. That is amplification of $1.7428\times$ and $3.1567\times$, matching the published $1.74\times$ and $3.16\times$ exactly. This check covers the monoculture case $R = \mathbf{11}^\top$, so it supports the base law the paper builds on, not any of the four new results. It is also our own earlier work, so it is not independent validation.

The linear prediction at full spillover is 2 and 3. The market comes in 12.9% below it at $N = 2$ and 5.2% above at $N = 3$. The two gaps have opposite signs, which is what a nonlinear market with saturating flow does around a linearization. The reference environment hits the prediction exactly because it leaves out that nonlinearity. The spread between the two modes, 3.4×10^{-3} against a theoretical zero, says the instability is purely common-mode, which is the mechanism Theorem 1 generalizes.

B. Why the rest are dry runs

We built a version of the order-flow simulator in which each dealer has its own response profile. It is exact when the profiles are flat, but it failed its acceptance test. Liquidity and price impact stay shared however the profiles differ, and the leftover error is larger than several of the effects the other checks resolve. We drew no figure from it and upgraded no check. The over-adaptation check is a dry run for a different reason: the simulator has no aggressiveness choice and no welfare measure. In the reference environment the over-adaptation gap grows from 29.0% at $N = 2$ to 119.9% at $N = 20$, and is exactly zero at $N = 1$, as Corollary 2 predicts.

VIII. LIMITATIONS AND CONCLUSION

A. What a supervisor can see

Every quantity above is hidden from a supervisor, but near the boundary the market gives itself away. As m_N approaches one, the leading principal component of public quotes has lag-one autocorrelation approaching one in magnitude, and its share of variance grows like $1/(1 - m_N)$. A supervisor can therefore estimate the distance to instability from public prices alone, and the estimate gets sharper as the market gets closer to the edge. Its consistency and sample complexity are open.

B. Measuring crowding on real models

The most useful next step is to put a number on $\lambda_{\max}(R)$ for real deployed models. Per-item outputs from many models on shared benchmarks, such as the correlated-errors data of [4], give a direct proxy: estimate the correlation matrix of per-item errors across models, take its leading eigenvalue, and read off N_{eff} for a market built on those models. The same data would show how far a mean-similarity index understates it. Error correlation is only a proxy for alignment of response directions, and closing that gap is part of the work.

C. Limitations

The analysis is linearized around equilibrium, and the measured check shows what that costs in a real market. The results with heterogeneous response directions are checked against the model, not measured, which is the largest gap in the work. Theorem 1 is exact for equal moduli and a two-sided bound otherwise, retraining is synchronous, and the clean $1 - 1/R_0$ form holds only in the $\kappa = s = 1$ corner. Measuring instability by stationary variance is a modeling choice. Unequal clocks, three or more correction levels, and corrections that change a firm's direction rather than its gain are named in the repository and not attempted.

D. Conclusion

Testing one model at a time cannot certify a market of models. The number that separates the two questions is effective crowding, and no test of a single model can see it. Current rules for general-purpose models tie their thresholds to model scale. We offer a market-level quantity instead, two remedies that substitute for each other, and a price.

REFERENCES

- [1] N. Beale, D. G. Rand, H. Battey, K. Croxson, R. M. May, and M. A. Nowak, "Individual versus systemic risk and the regulator's dilemma," *Proc. Natl. Acad. Sci. USA*, vol. 108, no. 31, pp. 12647–12652, 2011.
- [2] W. Wagner, "Diversification at financial institutions and systemic crises," *J. Financ. Intermed.*, vol. 19, no. 3, pp. 373–386, 2010.
- [3] A. E. Khandani and A. W. Lo, "What happened to the quants in August 2007?" *J. Invest. Manag.*, vol. 5, no. 4, pp. 29–78, 2007.
- [4] E. Kim, A. Garg, K. Peng, and N. Garg, "Correlated errors in large language models," in *Proc. ICML*, 2025.
- [5] J. C. Perdomo, T. Zrnic, C. Mandler-Dünner, and M. Hardt, "Performative prediction," in *Proc. ICML*, 2020.
- [6] R. M. Anderson and R. M. May, "Vaccination and herd immunity to infectious diseases," *Nature*, vol. 318, no. 6044, pp. 323–329, 1985.
- [7] P. Fine, K. Eames, and D. L. Heymann, "'Herd immunity': A rough guide," *Clin. Infect. Dis.*, vol. 52, no. 7, pp. 911–916, 2011.
- [8] V. Nagarajan and S. Ashok, "PRICE: Performative risk in coupled environments, code, derivations and certificates," <https://github.com/vignesh-nagarajan-vn/PRICE>, 2026.
- [9] —, "REFLEX: Reflexive equilibrium fixed-point learning for endogenous exchanges," arXiv:2608.16155, 2026.
- [10] M. Granovetter, "Threshold models of collective behavior," *Amer. J. Sociol.*, vol. 83, no. 6, pp. 1420–1443, 1978.
- [11] S. Bikhchandani, D. Hirshleifer, and I. Welch, "A theory of fads, fashion, custom, and cultural change as informational cascades," *J. Polit. Econ.*, vol. 100, no. 5, pp. 992–1026, 1992.
- [12] O. Diekmann, J. A. P. Heesterbeek, and J. A. J. Metz, "On the definition and the computation of the basic reproduction ratio R_0 in models for infectious diseases in heterogeneous populations," *J. Math. Biol.*, vol. 28, no. 4, pp. 365–382, 1990.
- [13] R. Pastor-Satorras and A. Vespignani, "Epidemic spreading in scale-free networks," *Phys. Rev. Lett.*, vol. 86, no. 14, pp. 3200–3203, 2001.
- [14] J. Kleinberg and M. Raghavan, "Algorithmic monoculture and social welfare," *Proc. Natl. Acad. Sci. USA*, vol. 118, no. 22, p. e2018340118, 2021.
- [15] R. Bommasani, K. A. Creel, A. Kumar, D. Jurafsky, and P. Liang, "Picking on the same person: Does algorithmic monoculture lead to outcome homogenization?" in *Proc. NeurIPS*, 2022.
- [16] K. Peng and N. Garg, "Monoculture in matching markets," in *Proc. NeurIPS*, 2024.
- [17] A. Narang, E. Faulkner, D. Drusvyatskiy, M. Fazel, and L. J. Ratliff, "Multiplayer performative prediction: Learning in decision-dependent games," *J. Mach. Learn. Res.*, vol. 24, 2023.
- [18] G. Piliouras and F.-Y. Yu, "Multi-agent performative prediction: From global stability and optimality to chaos," in *Proc. ACM Conf. Econ. Comput. (EC)*, 2023.
- [19] C. Mandler-Dünner, J. C. Perdomo, T. Zrnic, and M. Hardt, "Stochastic optimization for performative prediction," in *Proc. NeurIPS*, 2020.
- [20] Z. Izzo, L. Ying, and J. Zou, "How to learn when data reacts to your model: Performative gradient descent," in *Proc. ICML*, 2021.
- [21] J. P. Miller, J. C. Perdomo, and T. Zrnic, "Outside the echo chamber: Optimizing the performative risk," in *Proc. ICML*, 2021.
- [22] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, "Noise dressing of financial correlation matrices," *Phys. Rev. Lett.*, vol. 83, no. 7, pp. 1467–1470, 1999.
- [23] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, T. Guhr, and H. E. Stanley, "Random matrix approach to cross correlations in financial data," *Phys. Rev. E*, vol. 65, no. 6, p. 066126, 2002.
- [24] V. Nagarajan and A. Mittal, "PEBSA: Predicting economic behavior via sentiment analysis," *Int. J. Eng. Comput. Sci.*, vol. 13, no. 12, pp. 26656–26676, 2024.
- [25] J. M. Buchanan and W. C. Stubblebine, "Externality," *Economica*, vol. 29, no. 116, pp. 371–384, 1962.
- [26] A. O. Hirschman, "The paternity of an index," *Amer. Econ. Rev.*, vol. 54, no. 5, pp. 761–762, 1964.
- [27] T. Bergstrom, L. Blume, and H. Varian, "On the private provision of public goods," *J. Public Econ.*, vol. 29, no. 1, pp. 25–49, 1986.
- [28] M. L. Weitzman, "Prices vs. quantities," *Rev. Econ. Stud.*, vol. 41, no. 4, pp. 477–491, 1974.